HDOC DAY 59

PROBLEM

Write a C program that reads a string s. Pass the string to a function named longest_unique_substring() using its address. Use pointers to pass the string between the calling function and longest_unique_substring(). The function longest_unique_substring() must:

- Find the longest substring of s that contains no repeated characters.
- Calculate and display the length of the longest substring.
- Treat uppercase and lowercase letters as different characters.
- Display 0 if the input string is empty.

CODE

```
#include <stdio.h>
#include <string.h>

void longest_unique_substring(char *s)
{
   int i, j, start = 0, max_length = 0;
   int last_seen[256];

   for (i = 0; i < 256; i++)
      last_seen[i] = -1;

   for (i = 0; s[i] != '\0'; i++)
   {
      unsigned char ch = (unsigned char)s[i];
```

```
        if (last_seen[ch] >= start)
            start = last_seen[ch] + 1;

        last_seen[ch] = i;

        j = i - start + 1;

        if (j > max_length)
            max_length = j;
    }

    printf("Length of longest unique substring: %d\n",
           max_length);
}

int main()
{
    char s[1000];

    printf("Enter a string: ");
    if (fgets(s, sizeof(s), stdin) == NULL)
        return 0;

    s[strcspn(s, "\n")] = '\0';

    longest_unique_substring(s);

    return 0;
```

```
}
```